

Unit 2 - Fractions, Percentages, and Decimals

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The following B.E.S.T. Standards will be covered in this unit:

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals and percentages.

- *Clarification 1: Rational numbers include decimal equivalence up to the thousandths place.*

MA.6.NSO.2.3 Solve multi-step real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

- *Clarification 1: Within this benchmark, it is not the expectation to include both decimals and fractions within a single problem.*

Unit 2, Lesson 1: Equivalent Conversions: Fractions ↔ Decimals; Decimals ↔ Percentages



Benchmark

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.



Learning Targets

- ☐ I can convert between fractions and decimals.
- ☐ I can convert between percentages and decimals.



2.1.1 Warm-Up: Self-Evaluation

- For each problem, rate your level of confidence in your ability to answer each question, but don't actually answer them.

Use a scale from 1 to 5, where 1 means you do not feel confident about how to answer the question, and 5 means you feel very confident.

Problem	Confidence Level
Rewrite $2\frac{2}{5}$ in its equivalent decimal form.	
Rewrite 0.625 as a fraction with the same value.	
Rewrite 0.9 as an equivalent percentage.	
Rewrite 75% in its equivalent decimal form.	



2.1.2 Guided Instruction: Fraction and Decimal Conversions

Decimal Place Value Chart							
Thousands	Hundreds	Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths



The place value of a number provides the numerical value that a digit has by virtue of its position in a number.

Example: In 278.493

the 8 is in the **ones** place

the 4 is in the **tenths** place

the 9 is in the **hundredths** place

the 3 is in the **thousandths** place

1. Use the original number to fill in the missing digits and place values.

Original Number	Digit	Place Value
422.54	5	Tenths
5.276		Ones
12.749	4	
3.826		Thousandths

2. Consider the number 0.08.

- a. In the number 0.08, what is the place value of the final digit?
- b. What is the written description of the fraction $\frac{8}{100}$?
- c. Discuss with your partner any similarities or differences between the decimal 0.08 and the fraction $\frac{8}{100}$.



When rewriting a decimal that is greater than 0 but less than 1 as an equivalent fraction, use the place value of the final digit to determine the denominator of the fraction.

3. In the decimal 0.7, the final digit is in the

- ☐ tenths
- ☐ hundredths
- ☐ thousandths

place, so

$$\begin{array}{l} \circ \frac{7}{10} \\ \circ \frac{7}{100} \\ \circ \frac{7}{1000} \end{array}$$

is the equivalent

fraction form.

4. In the decimal 0.037, the final digit is in the

- ☐ tenths
- ☐ hundredths
- ☐ thousandths

place, so

$$\begin{array}{l} \circ \frac{37}{10} \\ \circ \frac{37}{100} \\ \circ \frac{37}{1000} \end{array}$$

is the equivalent

fraction form.

A fraction can be converted into its equivalent decimal form by dividing the **numerator** by the **denominator**.

5. To convert the fraction $\frac{1}{4}$ to a decimal, divide 1 by 4 by completing the diagram below.

4	1.00
Number of ones groups of 4	
Amount that remains	
Number of tenths groups of 4	
Amount that remains	
Number of hundredths groups of 4	
Amount that remains	





2.1.3 Guided Practice: Fraction and Decimal Conversions

1. Complete the table by converting each decimal to an equivalent fraction. If possible, write the equivalent fraction so that the numerator and denominator have no common factors.

Decimal Form	Fraction Form
0.08	$\frac{2}{25}$
0.27	
2.2	
0.005	

2. Complete the table by rewriting each fraction as its equivalent decimal.

Fraction Form	Your Work	Decimal Form
$\frac{1}{5}$		
$\frac{5}{8}$		
$\frac{7}{200}$		



2.1.4 Guided Instruction: Decimal and Percentage Conversions

A **percentage** is a ratio expressed as a fraction with a denominator of 100.

Percentages can also be written as fractions with a denominator of 100, or as a decimal.

- Write 66% as a fraction and as a decimal.

Percent	Fraction	Decimal
66%	$\frac{\quad}{100}$	

To convert a decimal to a percent, multiply the decimal by 100.

- Convert the decimal 0.16 to a percent.

$$0.16 \times \frac{\quad}{\quad} = \frac{\quad}{\quad} \%$$

To convert a percent to a decimal, divide the percent by 100 or multiply by $\frac{1}{100}$.

- Convert 16.2% to a decimal.

$$16.2 \div \frac{\quad}{\quad} \text{ or } 16.2 \cdot \frac{\quad}{\quad}$$



2.1.5 Guided Practice: Decimal and Percentage Conversions

- Complete the table by converting each decimal to an equivalent percentage.

Decimal Form	Percent Form
0.08	8%
0.27	
2.2	
0.005	

- Rewrite each percentage into its equivalent decimal form to complete the table.

Percent Form	Decimal Form
72%	0.72
12.5%	
0.7%	
214%	



2.1.6 Your Turn!

1. Rewrite each percentage as a decimal.

a. 125.6%

b. 3%

2. Convert each decimal to an equivalent percentage.

a. 0.008

b. 4.71

3. Rewrite each fraction as a decimal.

a. $\frac{1}{2}$

b. $2\frac{1}{10}$

4. Rewrite each decimal as an equivalent fraction.
Simplify where applicable.

a. 0.94

b. 1.7



2.1.7 Wrap-Up: Ticket Out the Door

1. Circle the expression that does not belong.

$\frac{2}{5}$	0.4
40%	$\frac{4}{8}$

2. Explain your reasoning for your choice.

Unit 2, Lesson 2: Equivalent Conversions: Fractions ↔ Percentages – Part 1



Benchmark

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.



Learning Targets

- ☐ I can convert between fractions with denominators of 100 and percentages.
- ☐ I can convert between fractions and percentages using equivalent fractions.



2.2.1 Warm-Up: A Brief History of Numerical Systems

Humans went from using body parts, tally marks, and symbols to count objects, to what we now know as the base-ten system. Multiple civilizations around the world developed positional numbering systems, and both Mayan and Indian civilizations developed the concept of “zero.” The base-ten system was perfected by eighth century Indian mathematicians. They realized that they could combine the concept of “zero” with a positional system that used repeatable symbols, whose values are determined by their position in a sequence. A base of ten was selected because, quite simply, we have ten fingers. Centuries later, this Hindu-Arabic numeral system began to spread to Europe and replaced the Roman numeral system. This is what we use today.



1. What is the historical importance of the number zero?

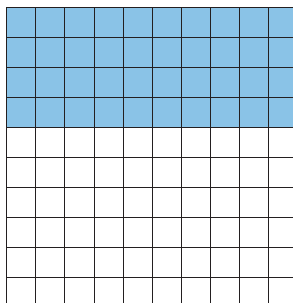
2. If zero was never created, how would that affect how we do math today?





2.2.2 Guided Instruction: Converting Between Fractions and Percentages

1. The model represents a numerical value. Use the model to answer the questions that follow.



- How many total squares are in the image?
- How many of the squares are shaded blue?
- What numerical value does the model represent, expressed as a fraction of $\frac{\text{blue shaded squares}}{\text{total squares}}$?

Recall that a **percentage** is a ratio expressed as a fraction of 100.

Percent (%) means “per 100,” “for every 100,” or “out of 100.”

For example, 40% means “40 out of 100,” so 40% is equivalent to the fraction $\frac{40}{100}$.

2. What decimal value has the same value as 40% and $\frac{40}{100}$?

Equivalent Values		
Percent Form	Fraction Form	Decimal Form
40%	$\frac{40}{100}$	

3. With your partner, complete the table by rewriting each value in the equivalent form indicated.

Fraction Form	Percent Form
$\frac{50}{100}$	50%
$\frac{71}{100}$	
$\frac{2}{100}$	
	150%
	42%

When converting fractions to percentages, generate an equivalent fraction with a denominator of 100 when possible.

4. Complete the table.

Original Fraction	Equivalent Fraction with a Denominator of 100	Percentage
$\frac{6}{10}$	$\frac{6}{10} \times \frac{10}{10} = \frac{60}{100}$	60%
$\frac{23}{50}$	$\frac{23}{50} \times \frac{2}{2} = \frac{\quad}{100}$	____%
$\frac{1}{5}$	$\frac{1}{5} \times \frac{\quad}{\quad} = \frac{\quad}{100}$	____%



2.2.3 Guided Practice: Converting Between Fractions and Percentages

1. Complete the table by rewriting each value in the equivalent form indicated. Simplify where applicable.

Fraction Form	Percent Form
$\frac{20}{50}$	40%
$\frac{13}{25}$	
$\frac{9}{10}$	
	30%
	18%
$\frac{5}{4}$	
	200%



2.2.4 Your Turn!

1. Rewrite each fraction as an equivalent percentage.

a. $\frac{17}{100}$

b. $\frac{42}{50}$

c. $\frac{36}{25}$

d. $\frac{96}{100}$

e. $1\frac{9}{100}$

f. $\frac{24}{20}$

g. $\frac{1}{50}$

h. $\frac{4}{5}$

2. Convert each to a fraction. Simplify where applicable.

a. 90%

b. 167%

c. 86%

d. 33%

e. 226%

f. 29%

g. 50%

h. 18%



2.2.5 Wrap-Up: Ticket Out the Door

1. Use the numbers below to create three more fractions equivalent to any three percentages in the table.

15	4	21	30	25	40
32	1	8	2	6	10
7	50	72	5	80	3

10%	20%	30%	40%	50%	60%	70%	80%	90%
								$\frac{72}{80}$

Unit 2, Lesson 3: Equivalent Conversions: Fractions ↔ Percentages – Part 2



Benchmark

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.



Learning Targets

- ☐ I can convert between fractions and percentages.
- ☐ I can rewrite positive rational numbers in different but equivalent forms.



2.3.1 Warm-Up: Matching Puzzle

Several values, expressed as decimals and percentages, are given.

0.71
7.1
0.071

710%
7.1%
71%

1. Match the values in the left column with their equivalent form in the right column by drawing lines to connect them.
2. Describe how you determined which values are equivalent.



2.3.2 Guided Instruction: Rewriting Rational Numbers in Equivalent Forms

Recall from Lesson 1 that a fraction can be converted into its equivalent decimal form by dividing the **numerator** by the **denominator**.

1. Complete the table by rewriting the fraction into an equivalent decimal form.

Fraction Form	Decimal Form
$\frac{13}{40}$	
$\frac{7}{8}$	
$2\frac{6}{10}$	



2. Complete the table by rewriting the decimal into an equivalent percentage.

Decimal Form	Percent Form
0.325	
0.875	
2.6	

3. What is the equivalent decimal form of $\frac{5}{8}$?

4. What is 0.625 rewritten as a percentage?

5. What is $\frac{5}{8}$ rewritten as a percentage?

6. When rewriting a fraction as a percentage, why might it be helpful to first rewrite it in its equivalent decimal form?

7. Complete the table by rewriting each value in its equivalent forms.

Fraction Form	Decimal Form	Percent Form
		67.5%
	0.019	
$\frac{63}{20}$		





2.3.3 Collaboration: Strategies for Rewriting Rational Numbers

Working with your partner, use the words in the bank to complete each statement. Words can be used more than once.

Word Bank

decimal denominator fraction
numerator percentage place value

1. To convert a _____ to a decimal, divide the _____ by the _____.
2. Multiply the _____ by 100 to convert a _____ to a _____.
3. To rewrite a _____ as a _____, divide the _____ by 100.
4. Use the _____ of the final digit to determine the denominator of the decimal when rewriting a _____ as an equivalent _____.
5. An equivalent fraction with a _____ of 100 can be used to convert a _____ to a percentage.



2.3.4 Guided Practice: Rewriting Rational Numbers in Equivalent Forms

1. Complete the table by rewriting each value in its equivalent forms.

Fraction Form	Decimal Form	Percent Form
$\frac{3}{8}$	0.375	37.5%
	0.22	
		4%
$2\frac{33}{50}$		



2.3.5 Wrap-Up: Ticket Out the Door

1. Rewrite each value as an equivalent percentage.

a. 0.02

b. $\frac{1}{5}$

c. $\frac{11}{8}$

2. Rewrite each value as an equivalent fraction.

a. 67%

b. 0.375

c. 35.65

Unit 2, Lesson 4: Using Operations to Solve Real-World Problems – Part 1



Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.



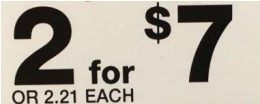
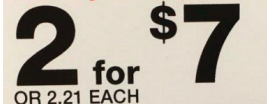
Learning Target

- ☐ I can determine which operation should be used to solve a real-world problem.



2.4.1 Warm-Up: Math Flop

- Determine the mistake in each picture. Then, complete the table.

Math Flop	The Mistake: Which number is not correct?	My Work: Show the math that proves it is a mistake.	The Correct Label: Fix the label by marking out the mistake and inserting the correct number.
			
			



2.4.2 Guided Instruction: Interpreting Real-World Problems

- Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in.). The length of the shorter board is $14\frac{5}{12}$ in.

Read each question and complete the following table.

- Which mathematical operation can be used to find the total length of the two boards?

Operation	Sketch a picture or explain why you chose that operation.

- Which mathematical operation can be used to find the length of each piece when the longest board is cut into four equal-sized pieces?

Operation	Sketch a picture or explain why you chose that operation.

- c. Check your answers with your partner and resolve any differences if necessary.
- d. Which mathematical operation can be used to determine the difference in the board lengths?
- e. Which mathematical operation can be used to find the total length of 48 of the shortest boards lined up end-to-end?

- f. Write the numeric expression that shows how to calculate the answer. Then, determine the units of the answer.

Information	Numeric Expression	Units of the Answer
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the total length of the two boards?		
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the length of each piece when the longest board is cut into four equal-sized pieces?		
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the difference between the board lengths?		
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the total length of 48 of the shortest boards lined up end-to-end?		





2.4.3 Guided Practice: Operations in Real-World Context

1. The table contains four different real-world contexts. Select the question in the middle column that requires the use of the operation indicated in the last column within this context.

Real-World Context	Which Question?	Operation
A Valencia orange tree that is twelve years old can produce 1.4 boxes of oranges. In May 2017, a box of fresh Valencia oranges cost \$20.45.	- For how many years did a box of Valencia oranges cost \$20.45? - How much money would a 12-year old Valencia orange tree generate? - What is the price per box of Valencia oranges?	$\times$
In 2016, there were 19.3 million Snapchat users in the age range of 18-24, and 16.7 million users in the age range of 25-34.	- How many Snapchat users were between the ages of 18 and 34? - What was the difference between the number of Snapchat users in the 18-24 and the 25-34 age range? - How many Snapchat users were younger than 18?	$+$

In 2004, each person in the United States consumed 61.5 pounds of refined sugar on average. In 2008, the average consumed per person was 66.3 pounds of refined sugar.	- What is the total amount of sugar the average person consumed in 2004 and 2008? - What was the average yearly increase in sugar consumption from 2004 to 2008? - What is the difference between the amount of sugar consumed in 2004 and 2008?	$-$
Barbara sent letters that exceeded the allowed weight by the U.S. Postal Service. To mail the letters she had to add a \$0.15 stamp to each letter. She spent \$2.55 in additional weight charges.	- What was the total cost of sending the letters? - For how many letters did Barbara pay the additional weight charge? - What is the difference between the cost of sending a regular letter and one that exceeds the weight allowed?	$\div$

2. Use the same four real-world contexts and the operation indicated to find the answer to the question selected in the previous table. Write the numeric expression used to determine the answer. Then, show your work and write the answer using the correct units.

Information	Numeric expression	Show your work and answer with units
A Valencia orange tree that is 12 years old can produce 1.4 boxes of oranges. In May 2017, a box of fresh Valencia oranges cost \$20.45.		
In 2016, there were 19.3 million Snapchat users in the age range of 18-24 and 16.7 million users in the age range of 25-34.		

In 2004, each person in the United States consumed 61.5 pounds of refined sugar on average. In 2008, the average consumed per person was 66.3 pounds of refined sugar.

Barbara sent letters that exceeded the allowed weight by the U.S. Postal Service. To mail the letters, she had to add a \$0.15 stamp to each letter. She spent \$2.55 in additional weight charges.





2.4.4 Wrap-Up: Reflection

Mathematics is most useful in the real world. Determine the operation(s) needed to find the answer to each question. Then, mark how confident you are in your choice of operation(s) to be used.

1. Fitzgerald is selling chocolate bars to raise money for a trip. He has to sell a total of 4 boxes. The first day, he sold $2\frac{3}{4}$ boxes. How many more boxes does he have to sell?

The operation(s) that should be used is/are

_____.

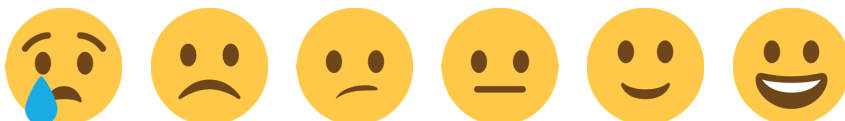


0 1 2 3 4 5 6 7 8 9 10

2. Ginger is buying new carpet for her living room. The carpet costs \$56.81 per square yard with an additional \$7.64 per square yard to install it. The size of the living room is 13.6 square yards. How much will the carpet for the living room cost?

The operation(s) that should be used is/are

_____.



0 1 2 3 4 5 6 7 8 9 10

Unit 2, Lesson 5: Using Operations to Solve Real-World Problems – Part 2



Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.



Learning Target

- ☐ I can solve real-world problems with up to two steps using decimals or fractions.



2.5.1 Warm-Up: Two Questions and One Wonder

Ky-ree, a firefighter in Los Angeles, budgets his paycheck.

Ky-ree's Budget	
Monthly salary	\$5,800
Taxes and other deductions	\$2,900
Net take home income	\$2,900
Rent	\$1,350
Utilities	\$60
Internet & cable	\$50
Cell phone	\$130
Car payment	\$330
Insurance	\$120
Gas	\$140
Food	\$250
Gym	\$30
Entertainment	\$120
Remaining	\$320

1. List two questions you have about Ky-ree's budget.
2. List one thing you wonder about Ky-ree's budget.



2.5.2 Guided Instruction: Real-World Contexts About Finances

Income is the amount of money received.

1. Recall the problem from Lesson 4.

A Valencia orange tree that is 12 years old can produce 1.4 boxes of fruit. In May 2017, a box of fresh Valencia oranges cost \$20.45. How much income would a Valencia orange tree generate?

Finance is the management of money.

2. Daniel owns a Valencia orange grove. Choose other items from the list that he should consider as a part of his finances.
 - ☐ The taxes he pays on the property.
 - ☐ The amount of money he has in the bank.
 - ☐ The cost of taking care of the orange grove.

3. Use the words in the word bank to complete each statement.

Word Bank

withdrawal checking costs deposit
expenses savings taxes

- a. _____ or _____ describe money that is spent.
- b. _____ are the amount of money that is paid to the government.
- c. A _____ is the amount of money placed in a bank account.
- d. A _____ is the amount of money removed from a bank account.
- e. _____ and _____ are types of bank accounts.

An **investment** is when the initial amount of money, known as the **principal**, earns more money, known as the **return**, over time.

4. For each financial transaction, write the mathematical operation that is most commonly associated with it.

Financial Transaction	Mathematical Operation
costs	
deposit	
expenses	
income	
taxes	
withdrawal	
return on an investment	

Gross income is the amount of income that is earned.

5. Hannah earns \$9.17 an hour. Each week she works 14 hours. How much is Hannah's weekly gross income?

Net income is the amount of income that is earned after expenses and taxes are deducted.

6. Each week Hannah has to pay \$26.96 in taxes. What is Hannah's net income?



2.5.3 Guided Practice: Solving Real-World Finance Problems

1. A trampoline park charges \$26.88 for 2 hours.
 - a. How much is the cost for 1 hour?
 - b. The manager of the trampoline park knows that during a Saturday there are typically 125 people at the park. If each person only pays for 2 hours, what is the park's typical income on Saturdays?
 - c. The manager of the trampoline park lists the costs for Saturday as follows: \$119.50 to pay the employees and \$256.73 for all other costs. What are the park's typical costs?
 - d. Determine the park's net income for a typical Saturday.
2. Emmanuel invested \$12,500 into the stock market. His return, the money he earned on his investment, was \$281.25 for each of the first three quarters of the year. He combines his returns with his initial investment. How much does Emmanuel have invested after the third quarter of the year?
3. Aaron has \$2,478.23 in his checking account. He writes a check for \$637.89 to pay for new tires on his truck. How much money does he have in his checking account now?
4. Violeta has \$5,734.49 in her savings account. She deposits \$218.56 and withdraws \$92.50. How much does she have in her savings account now?





2.5.4 Wrap-Up: Ticket Out the Door

Mishka is planning to host a music festival to raise funds for a local food bank. Her budget is shown.

Item	Amount
Ticket price at the door	\$9.00
Fee paid to each band	\$175.00
Early-bird ticket price	\$6.50
Equipment rental	\$225.00
Publicity (flyers/artwork)	\$64.57
Stage rental	\$187.50

1. In order to find the **gross income** for the music festival, I need to know _____ and _____.
2. List the items in Mishka's budget that are **expenses**.
3. If Mishka had a gross income of \$1,166.50 and expenses totaling \$827.07, what is her net income for the music festival?

Unit 2, Lesson 6: Solving Two-Step, Real-World Problems with Fractions or Decimals



Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.



Learning Target

- ☐ I can solve real-world problems with two steps using decimals or fractions.



2.6.1 Warm-Up: Would You Rather?

Would you rather buy 8 Skittles for 25 cents or 62 Skittles for \$1.49?



1. Explain your choice.

2. Would your answer change if you could choose what flavor the 8 Skittles were but you could not choose the flavors from the bag?



2.6.2 Guided Practice: Two-Step Rational Problems

1. Sophie grew her hair out for a long period of time. Her hair is $2\frac{3}{5}$ feet long. She donated her hair by cutting $\frac{5}{6}$ of a foot off. Sophie's hair then grew $1\frac{2}{3}$ feet.
 - a. What operations are needed to determine the length of Sophie's hair?

 - b. What values are necessary to determine the length of Sophie's hair?

 - c. Write the numeric expression that can be used to determine the length of Sophie's hair.

 - d. Determine the length of Sophie's hair.

 - e. Determine if the length of Sophie's hair makes sense. Support your answer.

2. Tiffany has a balance of \$167.32 in her checking account. She deposits \$58.16 in her checking account and then writes a check for \$97.27.
 - a. What operations are needed to determine the amount of money in Tiffany's bank account?
 - b. What values are necessary to determine the amount of money in Tiffany's bank account?
 - c. Write the numeric expression that can be used to determine the amount of money in Tiffany's bank account.
 - d. Determine the amount of money Tiffany has left in her checking account.



2.6.3 Collaboration: Error Analysis

1. Stefanie solved the following word problem. Her work is shown.
 - a. Evaluate her work and determine where she made mistakes.

Problem	Stefanie's Work
Janelle has $2\frac{1}{4}$ pans of brownies. Each pan made 8 brownies. She gave $\frac{2}{3}$ of the brownies to her friends. How many brownies did Janelle give away?	Step 1: $2\frac{1}{4} \times 8 = \frac{9}{4} \times 8 = \frac{18}{4}$ Step 2: $\frac{18}{4} \div \frac{2}{3} = \frac{18}{4} \times \frac{3}{2} = 6\frac{3}{4}$ Answer: She gave away $6\frac{3}{4}$ brownies.

- b. Help Stefanie understand and learn from her mistakes. Write a brief note to Stefanie explaining what her mistakes were and how she could learn from those mistakes.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

- c. Read your partner's note to Stefanie and offer suggestions for how to improve their note.





2.6.4 Your Turn!

1. Martha fills her ice cream bowl with $\frac{3}{4}$ cup of ice cream, $\frac{1}{8}$ cup of candy toppings, and the rest with fruit toppings. If the bowl contains a total of $1\frac{1}{2}$ cups, how much of the bowl is made up of fruit toppings?
2. Jasmine has a ribbon that is 12.6 meters long. Sharon has a ribbon that is 3.8 meters shorter. What is the total length of the two ribbons?
3. Before soccer practice, Hernando puts $4\frac{1}{8}$ cups of sports drink in a water bottle with $1\frac{2}{3}$ cups of ice. After a hot afternoon of practice, he drinks $2\frac{1}{5}$ cups of the sports drink and melted ice. How many cups of this mixture are left in the water bottle?
4. A bag of chocolates weighs 1.384 pounds. If Luis gives each of his 6 friends 0.23 pounds of chocolate, how many pounds of chocolate does Luis have left?





2.6.5 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of each of today's learning targets. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can solve real-world problems with two steps using decimals.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



2. I can solve real-world problems with two steps using fractions.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



Unit 2, Lesson 7: Solving Multi-Step, Real-World Problems with Fractions or Decimals



Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.



Learning Target

- ☐ I can solve real-world problems with three steps using decimals or fractions.



2.7.1 Warm-Up: Short Division

In Mexico, when dividing by a single-digit divisor, sometimes a method called “short division” is used. Consider the example shown of $74 \div 4$.

$$\begin{array}{r} 18 \\ 4 \overline{) 74} \end{array}$$

1. Using your knowledge of division, explain where you think the **1**, **3**, **8**, and **2** came from in the process.
2. Show how you might represent the same division problem using a method different from “short division.”



2.7.2 Guided Instruction: Three-Step Rational Problems

1. For his recital, a student practiced the piano 2.18 hours each day for 5 days last week. This week, he practiced 0.75 hours each day for 4 days. How many hours did he spend practicing?

2. The Across Florida 200 Trail Run is an event where participants run 200 miles of trails between the Withlacoochee Bay Trail and Crescent Beach. To prepare for the Across Florida 200 Trail Run, Walt ran 2.18 hours each day for 5 days last week. This week, he ran 2.75 hours each day for 4 days. Walt runs at an average pace of 6.23 miles per hour.

- How many hours has Walt run over the last two weeks?
- How many miles has Walt run over the last two weeks?
- Based on Walt's average pace, how much longer would it take him to run the Across Florida 200 Trail Run than the amount of time he has spent practicing the last two weeks?

3. The Across Florida 200 Trail Run offers a team competition where a team of 4 runners take turns to run the 200 miles.

A team of runners and their average pace is shown.

Name	Molly	Reginald	Tyler	Liz
Average Pace (miles per hour)	6.75	6.75	8.2	9.58

To prepare for the race the team ran part of the trail.

Team Member	Description	Miles
Tyler	Withlacoochee Bay Trail to Lake Rousseau Dam & Recreational Area	10
Liz	Lake Rousseau to Florida Trail	15
Molly	Florida Trail Part 1	35
Reginald	Florida Trail Part 2	25

How long did it take the team to run the practice run of 85 miles?



2.7.3 Collaboration: Distance Between Two Hurdles

There are 10 equally spaced hurdles on a racetrack. The first hurdle is 13.72 meters from the start line. The final hurdle is 14.02 meters from the finish line. The racetrack is 110 meters long.



The diagram below represents what the track would look like if the track were straight and not curved.

1. Use the diagram below to show the placement of the hurdles on the racetrack. Label all known measurements.



2. Discuss with your partner how many spaces there are between the 10 hurdles.
3. How far are the hurdles from one another? Explain or show your reasoning.

4. A professional runner takes 3 strides between each pair of hurdles. The runner leaves the ground 2.2 meters before the hurdle and returns to the ground 1 meter after the hurdle.
 - a. Draw a diagram to model the runner's strides between the hurdles. Label all known measurements.
 - b. About how long are each of the runner's strides between the hurdles? Show your reasoning.



2.7.4 Your Turn!

1. A pool that holds 273.68 gallons of water lost 4.79 gallons of water each day while there was a leak that went unnoticed. Tad noticed the water was lower, so he added 20.45 gallons of water before he realized there was a leak on the side. If Tad fixed the leak after 8 days of leaking, how many gallons of water does he currently have in the pool?
2. A grocery store opens a new wing bar that charges by the pound. Yasmine wants to purchase $\frac{1}{4}$ pound of garlic parmesan wings, $\frac{1}{8}$ pound of mild wings, and $\frac{2}{5}$ pound of teriyaki wings. She also wants to purchase some hot wings. If Yasmine wants to purchase a total of 2 pounds of wings, how many pounds of hot wings should she buy?



2.7.5 Wrap-Up: Cool Down

1. For each statement, write whether you agree with it, disagree with it, or are unsure.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	
When you add two positive decimals together, the value is always larger.	
To add fractions, their denominators must be the same.	
The product of two fractions is always a larger number.	
When you put a zero on the end of any number, it makes it larger.	